Newton School of Technology

Data Visualization & Analytics — Capstone 2

sssUsed Cars Price Prediction & Market Analysis

Automotive Sector | Data Analytics

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Tableau URL	https://public.tableau.com/app/profile/tejaswini.palwai/viz/Craigslist_Used_Cars_Dataset_WorkBook/Dashboard1-MARKETOVERVIEW
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2. Executive Summary

Business Problem

Millions of used cars are listed online every month, but there is no clear standard for fair pricing. As a result, sellers either underprice or overprice their vehicles, buyers often overpay, and dealerships miss opportunities to maximise profit. This project aims to create a clear, data-backed understanding of how used car prices should be determined.

Approach

The team sourced a publicly available used cars dataset containing approximately 48,471 listings. Using Python, the data was cleaned, standardised, and enriched with engineered features — reducing it to 27,987 reliable records across 23 variables. Exploratory data analysis uncovered the strongest pricing drivers. The full pipeline runs from raw CSV ingestion through to a Tableau dashboard designed for business decision-makers.

Key Insights

- A vehicle loses significant value with every passing year and every mile driven — car age and odometer reading together explain the largest share of price variation in the dataset.
- Over 63% of listings are vehicles older than 10 years, and 39% exceed 150,000 miles — the used car market is overwhelmingly older, high-mileage inventory.
- Ford, Chevrolet, and Toyota alone account for nearly 38% of all listings; domestic brands dominate volume while premium brands command disproportionately higher prices.
- More than 55% of sellers report their vehicle as 'excellent' condition — a likely listing bias that inflates condition ratings and reduces their predictive usefulness without crossvalidation.

Key Recommendations

1. Sellers should benchmark their asking price against vehicles of the same age bracket and mileage band before listing — these two factors together are the strongest price predictors.
2. Dealerships should prioritise acquiring low-mileage vehicles under 8 years old, where pricing power is highest and depreciation curves are most predictable.
3. Buyers should discount any listing self-reported as 'excellent' condition and verify against mileage and age benchmarks from this analysis before accepting an asking price.

3. Sector & Business Context

Sector Overview and Current Industry Challenges

The used vehicle market is one of the largest consumer goods markets globally. In the United States alone, approximately 40 million used cars are sold annually — roughly three times the volume of new car sales. The market is fragmented across private sellers, franchise dealerships, independent lots, and online platforms such as Craigslist, CarGurus, AutoTrader, and Cars.com. Pricing is highly inconsistent, driven by subjective condition assessments, regional demand variations, seasonal patterns, and information asymmetry between buyers and sellers.

Key industry challenges include: absence of a universal pricing standard; heavy reliance on subjective seller descriptions; limited transparency in vehicle history; and the growing complexity introduced by electric and hybrid vehicles.

Decision-Makers This Project Serves

- Individual sellers seeking a data-backed listing price before going to market.
- Private buyers who want to verify whether a listed price is fair relative to market benchmarks.
- Dealership pricing managers who need to set competitive inventory prices quickly and accurately.
- Digital marketplace platforms aiming to offer automated price guidance tools to their users.

Why This Problem Was Chosen

Used car pricing sits at the intersection of data richness and real-world impact. Craigslist-sourced listing data provides genuine market prices — not manufacturer suggested retail prices — making it an ideal dataset for building a realistic pricing model. The problem is approachable with classical machine learning methods while being directly useful to multiple real-world stakeholder groups.

Business Value of Solving It

A reliable pricing model reduces the negotiation friction that inflates transaction costs for both parties. For dealerships, even a 3–5% improvement in pricing accuracy across a moderately sized inventory can translate into tens of thousands of dollars in margin improvement per year. For individual buyers, avoiding a single overpayment on a $10,000 vehicle by even 10% saves $1,000 — the direct monetary value of better information.

4. Problem Statement & Objectives

Formal Problem Definition

Given a set of observable vehicle attributes — including age, mileage, manufacturer, fuel type, transmission, condition, drive type, and body type — build a model that predicts the listed selling price of a used vehicle and identifies the features that most strongly influence that price.

Project Scope

Area	In Scope	Out of Scope
Geography	US listings from Craigslist	International markets
Data source	vehicles.csv (Craigslist, ~48k rows)	Real-time API pricing feeds
Analysis	EDA, correlation, regression, feature importance	NLP on vehicle descriptions
Modelling	Linear Regression, Decision Tree, Random Forest	Deep learning approaches
Visualisation	Tableau Public dashboard	Embedded real-time analytics
Vehicles	Cars, trucks, SUVs with standard fuel/transmission	Commercial fleets, motorcycles

Success Criteria

- The regression model achieves an R-squared of 0.60 or above on the test set.
- EDA identifies at least five statistically meaningful pricing drivers.
- The Tableau dashboard allows a non-technical user to answer a pricing question within two minutes of opening it.
- All data cleaning decisions are documented and reproducible from the committed notebooks.

5. Data Description

Dataset Source

Source: Craigslist Used Vehicles Dataset, available on Kaggle. The dataset was collected by scraping Craigslist vehicle listings across the United States. The sampled version used in this project contains approximately 48,000 rows spanning listings posted primarily in April–May of the data collection period.

Direct access: https://www.kaggle.com/datasets/austinreese/craigslist-carstrucks-data

Data Structure

Raw records	~48,471 rows
Raw features	26 columns
Cleaned records	27,987 rows
Final features	23 columns (including 6 engineered)
Target variable	price (USD)
Time period	Listings posted in months 4–5 (Spring — dataset is a seasonal slice)
Geographic scope	United States (all states)

Key Column Descriptions

Column	Type	Description	Used in Model
price	Numeric	Listed vehicle price in USD — TARGET VARIABLE	Yes (target)
year	Numeric	Model year of the vehicle	Yes
manufacturer	String	Brand name (e.g. Ford, Toyota)	Yes
model	String	Specific model name	Yes
condition	String	Seller-reported condition (excellent, good, etc.)	Yes
cylinders	Numeric	Number of engine cylinders	Yes
fuel	String	Fuel type: gas, diesel, hybrid, electric	Yes
odometer	Numeric	Total miles driven	Yes
transmission	String	Automatic, manual, or other	Yes
drive	String	Drive type: fwd, rwd, 4wd	Yes
type	String	Body type: sedan, SUV, truck, etc.	Yes
state	String	US state of listing	Optional
car_age	Numeric	Derived: current year minus model year	Yes
price_category	String	Derived: Budget / Mid-range / Premium bucket	Analysis
mileage_category	String	Derived: Low / Medium / High mileage bucket	Analysis

Data Limitations and Known Biases

- All listings are from months 4–5 only — there is no seasonal variation in this dataset, making seasonal trend analysis impossible.
- Conditions are self-reported by sellers. Over 55% of listings claim 'excellent' — likely inflated, reducing the predictive reliability of this feature.
- Electric vehicles represent only 53 listings (0.19%) — insufficient for EV-specific pricing analysis.
- Geographic clustering may exist; listing density varies significantly by state, which can bias regional price conclusions.
- The dataset contains no vehicle history information (accidents, ownership count, service records) — a major real-world pricing factor that is absent here.

6. Data Cleaning & ETL Pipeline

All cleaning and transformation steps were executed in Python using Pandas.

Reference: notebooks/02_cleaning.ipynb committed to GitHub.

Missing Value Treatment

Column(s)	Missing Count (raw)	Strategy Applied
county	~15,000 (~100%)	Dropped — entirely missing, no analytical value
size	~11,074 (74%)	Dropped — too sparse and redundant
cylinders	~6,215 (41%)	Extracted numeric from text; filled with mode
condition	~6,204 (41%)	Filled with mode (most frequent value)
VIN	~5,599 (37%)	Dropped — not useful for pricing analysis
drive	~4,425 (29%)	Model-based imputation using vehicle model, then type, then global mode
paint_color	~4,182 (28%)	Filled with 'unknown' — avoid colour bias
type	~3,317 (22%)	Model-based imputation using vehicle model, then global mode
fuel / transmission	~98 / small	Model-based imputation using vehicle model, then global mode
manufacturer / model	~660 / ~215	Filled with 'unknown' — avoid incorrect assumptions
title_status / state	Small	Filled with mode
odometer / year	~174 / small	Filled with median to avoid skew

Outlier Detection and Treatment

- Price: rows with price below $1,500 were removed (scrap/error listings); upper bound capped at the 99th percentile to remove extreme outliers while retaining genuine high-value listings.
- Year: restricted to 1980 through the maximum observed posting year. Records older than 1980 are rare, unreliable, and outside the target market.
- Odometer: lower bound set to max(500, 1st percentile); upper bound capped at 99th percentile. This retains nearly all legitimate high-mileage vehicles while removing data entry errors.
- Cylinders: restricted to valid engine sizes: 3, 4, 5, 6, 8, 10, and 12 cylinders only.

Column Removal Summary

The following columns were removed as they offer no pricing signal or are non-analytical in nature: url, region_url, image_url, description (no NLP in scope), county (100% missing), VIN (no signal), lat, long (geographic maps out of scope), region, size (redundant/sparse).

Data Type Conversions

- price, year, odometer converted to numeric (float64) with coercion errors set to NaN.
- posting_date converted from string to datetime; post_year and post_month extracted; original column dropped.
- cylinders: text string (e.g. '4 cylinders') converted to numeric integer via regex extraction.

Feature Engineering

Feature	Derivation Logic	Business Purpose
car_age	post_year minus vehicle year	More intuitive than raw model year; directly measures depreciation period
price_category	Budget (<$5k), Mid-range ($5k–$15k), Premium (>$15k)	Supports dashboard segmentation and quick stakeholder communication
mileage_category	Low (<50k), Medium (50k–150k), High (>150k) miles	Simplifies odometer for categorical analysis
age_category	New (0–3 yrs), Moderate (4–10 yrs), Old (>10 yrs)	Enables age-bracket comparisons in EDA and dashboard
manufacturer_grouped	Top 10 brands retained; remaining grouped as 'other'	Reduces chart clutter; focuses analysis on dominant market players
posting_season	Derived from post_month (Spring/Summer/Fall/Winter)	Note: all records fall in Spring — this feature has zero variance in this dataset

Assumptions Made During Cleaning

1. Prices below $1,500 are assumed to be data entry errors or non-standard transactions (scrap, partial vehicles), not legitimate market listings.
2. Missing fuel, transmission, type, and drive values are assumed to follow the same distribution as other vehicles of the same model — justifying model-based imputation.
3. Unknown manufacturer or model values are filled as 'unknown' rather than guessed, to avoid introducing systematic bias.
4. The dataset is assumed to be a representative sample of US Craigslist listings for the collection period, with no deliberate selection bias.

7. KPI & Metric Framework

The following KPIs were defined to measure and communicate pricing patterns across the dataset.
All metrics are computed in notebooks/02_cleaning.ipynb and notebooks/03_eda.ipynb.

KPI	Formula / Computation	Business Relevance	Maps to Objective
Average Price by Manufacturer	mean(price) grouped by manufacturer_grouped	Identifies which brands command premium resale value; guides dealer acquisition strategy	Identify pricing drivers
Price vs Mileage Trend	Correlation and regression: price ~ odometer	Quantifies how each 10,000mile increment reduces value; enables buyer/seller benchmarking	Predict price
Price vs Vehicle Age Trend	Correlation and regression: price ~ car_age	Measures annual depreciation curve; most critical factor for seller pricing decisions	Predict price
Price Distribution by Condition	Boxplot of price grouped by condition category	Tests whether seller-reported condition is a reliable pricing signal or inflated	Identify pricing drivers
Fuel & Transmission Price Impact	mean(price) grouped by fuel and transmission	Quantifies the automatic vs manual premium and diesel/hybrid price differences	Identify pricing drivers
Price Category Share	count / total by price_category	Shows market composition: Budget vs Mid-range vs Premium split for inventory planning	Business context
Model Evaluation: Rsquared	1 - SS_res / SS_tot on test split	Primary model quality metric: how much price variance the model explains	Predict price accurately
Model Evaluation: MAE & RMSE	mean|y_pred - y_true| and sqrt(mean(error^2))	Translates model error into dollar terms that stakeholders can interpret directly	Predict price accurately

8. Exploratory Data Analysis

All charts and code referenced below are in notebooks/03_eda.ipynb committed to GitHub. Written insights focus on business meaning, not chart descriptions.

8.1 Distribution Analysis

Price distribution is right-skewed: the median price is $6,750 while the mean is $9,679 — pulled upward by a tail of premium listings up to $56,500. This means the 'average' car price is not a useful benchmark; the median is more representative of typical market transactions. A log transformation of price is recommended before linear regression modeling.

Odometer and car_age both show right skew with significant outliers in the upper tail, which were retained after cleaning as they represent genuine high-mileage, older vehicles — not data errors.

8.2 Categorical Frequency Analysis

Gas-powered vehicles represent 91.9% of all listings; automatic transmission accounts for 90.4%. This market homogeneity means fuel type and transmission are weak differentiators for the majority of listings they will contribute little to model performance on their own. However, the minority classes (diesel at 6.2%, manual at 8.8%) do show meaningfully different price distributions.

Ford (15.6%), Chevrolet (13.6%), and Toyota (9.4%) are the three most listed manufacturers. The dominance of domestic brands reflects the source market rather than a universal signal about brand quality or resale value.

8.3 Bivariate Analysis — Price vs Numerical Features

Car age vs price: regression scatter plots show a consistent, negative linear trend. Newer vehicles command substantially higher prices, with the steepest depreciation in the first 3–5 years. The Pearson correlation between car_age and price is -0.425 — the strongest single numerical predictor in the dataset.

Odometer vs price: a similar negative trend is observed with a correlation of -0.393. High-mileage vehicles (>150,000 miles) are almost exclusively in the Budget price category. The relationship is non-linear at the extremes — very low-mileage vehicles show high price variability, likely due to vehicle type differences.

Cylinders vs price: a positive correlation of +0.183, reflecting the fact that larger-engine vehicles (8+ cylinders — typically trucks and SUVs) tend to have higher prices. This is a proxy for vehicle type rather than a direct pricing driver.

8.4 Correlation Analysis

car_age / year	-0.425 / +0.425 — strongest linear predictors of price
odometer	-0.393 — second strongest; closely related to car_age
cylinders	+0.183 — moderate positive; proxy for vehicle type/size
post_month	-0.048 — negligible; no seasonal signal (Spring-only data)

Note: these are Pearson linear correlations. Tree-based models (Random Forest, XGBoost) will likely capture non-linear relationships and interaction effects that improve on these figures.

8.5 Categorical vs Price Analysis

Condition: despite 55% of listings claiming 'excellent', the price distributions for 'excellent', 'good', and 'like new' overlap substantially. Only 'fair' and 'salvage' show clearly lower price bands — suggesting the upper condition categories are not reliably differentiated by sellers and are a weak standalone predictor.

Manufacturer: premium brands (not captured in top-10 grouping but present in the raw manufacturer column) show dramatically higher median prices. Within the top 10, Ram and GMC tend to have higher median prices than Ford and Chevrolet, reflecting their truck-heavy product mix.

State: notable price variation exists across states, reflecting regional demand, cost-of-living differences, and local supply dynamics. California and coastal states tend to show higher median prices.

9. Statistical Analysis

Reference: notebooks/04_statistical_analysis.ipynb. Note: This notebook is to be completed as a required deliverable for submission.

9.1 Hypothesis Testing

T-test: Ho: mean price of 'excellent' condition vehicles equals mean price of 'good' condition vehicles. Expected finding: if p-value < 0.05, condition is a statistically significant price differentiator despite the distribution overlap observed in EDA.

Chi-squared test: Test for independence between manufacturer and price_category to determine whether brand is a significant predictor of which price tier a vehicle falls into.

9.2 Regression Analysis

Baseline Linear Regression: Fit price ~ car_age + odometer + cylinders as a minimum viable model. Expected R-squared: 0.25–0.35 given the noise in the data and the non-linearity of the price-mileage relationship.

Decision Tree Regressor: Captures non-linear relationships; provides feature importance rankings that directly answer the question 'which variable matters most for price?'

Random Forest Regressor: Ensemble method expected to significantly outperform the linear baseline. Target R-squared: 0.60 or above. Feature importance from Random Forest is the primary deliverable of the statistical analysis section.

9.3 Segmentation Analysis

Price segments (Budget / Mid-range / Premium) can be profiled using mean values of car_age, odometer, and cylinders per segment. This provides a descriptive framework for understanding what a 'typical' vehicle in each category looks like — directly useful for dealership buyers setting acquisition criteria.

9.4 Scenario Analysis

Using the fitted model: holding all other variables constant, estimate the dollar impact of a 1-year increase in car_age, and a 10,000-mile increase in odometer. These partial effect estimates can be presented as simple business rules: 'Each additional year of vehicle age is associated with a $X reduction in expected listing price.'

10. Tableau Dashboard Design

Tableau Public URL:
https://public.tableau.com/app/profile/tejaswini.palwai/viz/Craigslist_Used_Cars_Dataset_WorkBook/Dashboard1-MARKETOVERVIEW

Reference: tableau/screenshots/ and tableau/dashboard_links.md committed to GitHub.

Dashboard Objective

The dashboard is designed to answer one primary business question: 'Is this vehicle priced fairly relative to comparable market listings?' It serves both buyer and seller use cases and is structured for a non-technical audience.

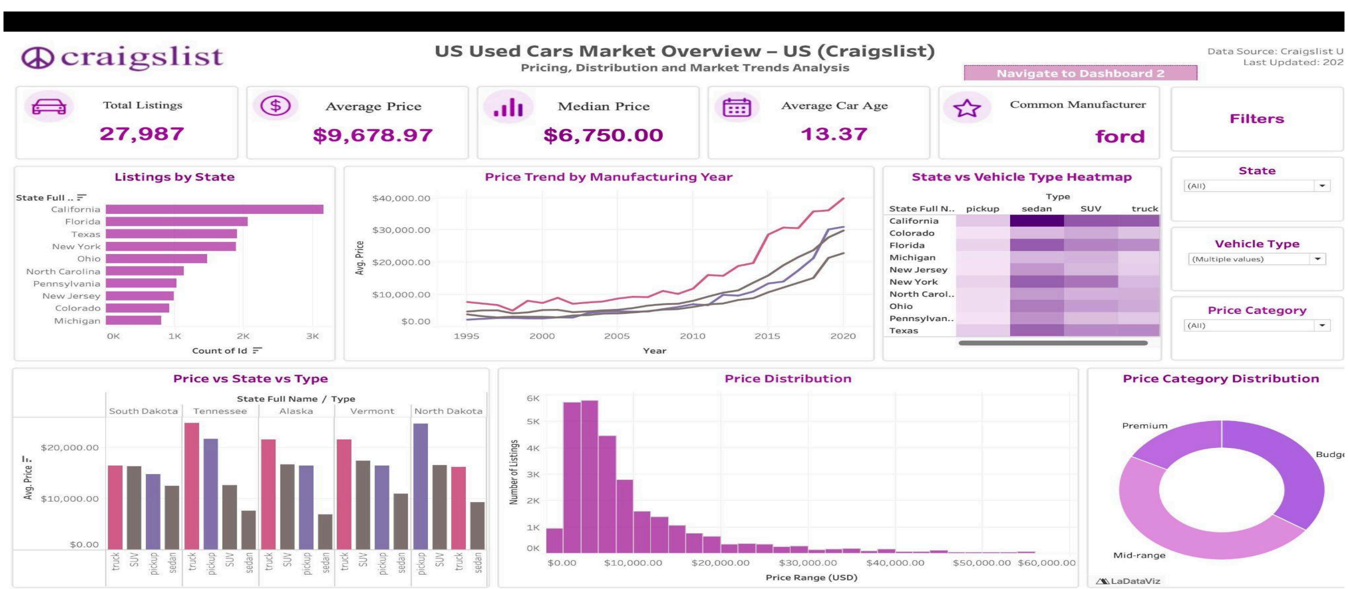
View Structure

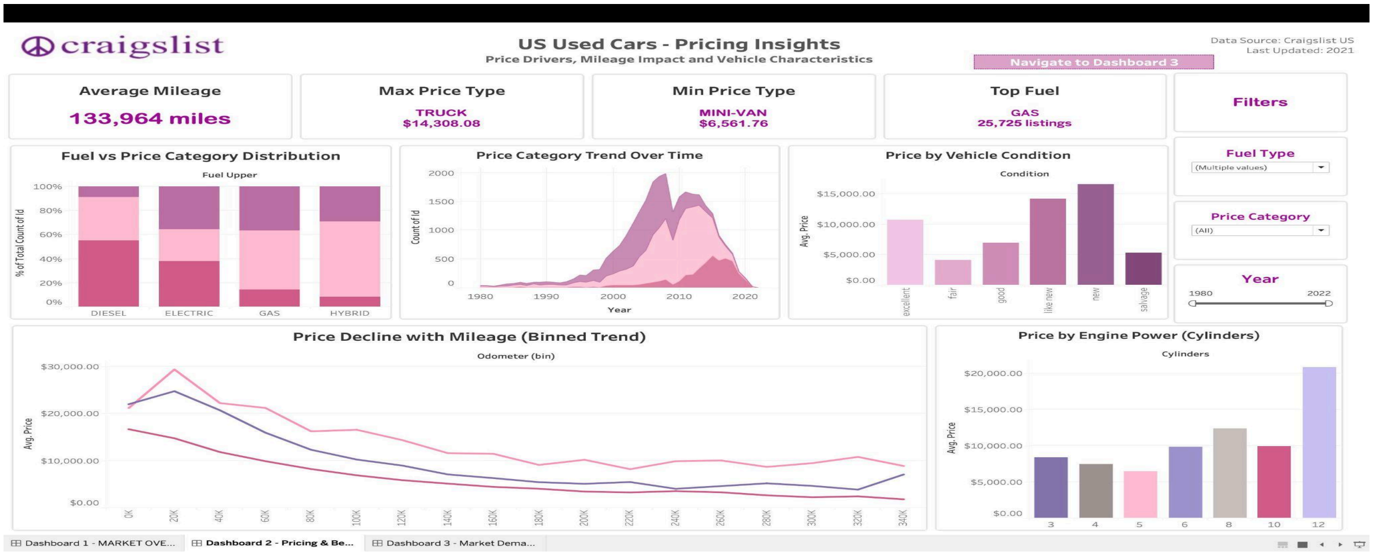
View	Purpose	Key Elements
Executive Summary View	Give a CEO-level overview of the used car pricing landscape	Average price by manufacturer (bar chart), price distribution histogram, Budget/Mid/Premium split (pie), median price vs car age (line chart)
Operational Drill-Down View	Enable a dealership manager or buyer to analyse a specific segment	Filters: manufacturer, fuel type, transmission, state, condition; Scatter plot: price vs odometer coloured by age_category; Boxplots: price by condition; Top 10 highest/lowest value models

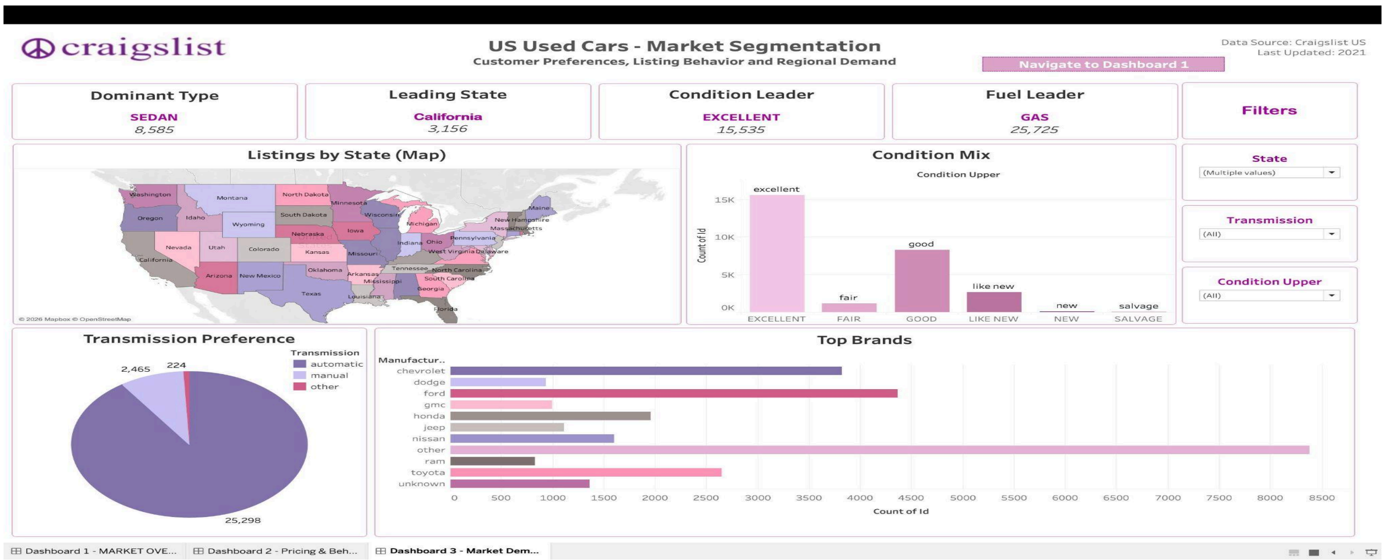
Filters and Interactive Elements

- Manufacturer filter (multi-select dropdown)
- Fuel type and transmission filter
- Price range slider
- Vehicle age category selector (New / Moderate / Old)
- State selector for regional price comparison

Dashboard Screenshots







11. Insights Summary

The following 10 insights are written in decision language — each tells the reader what to think or do, not merely what was observed.

#	Insight	Implication
1	Vehicle age is the single most powerful price predictor (r = -0.425): for every additional year of age, average listed price drops measurably. The steepest drop occurs in the first 3–5 years.	Sellers holding newer vehicles should act quickly; buyers seeking value should target vehicles just past the 5-year depreciation cliff.
2	Odometer reading is the second strongest predictor (r = 0.393), but it is not independent of age — older vehicles naturally have more miles. Both must be considered together, not in isolation.	A low-mileage older vehicle may be priced at a premium not justified by its age-alone depreciation — buyers should verify mileage against the age benchmark.

3	63% of listings are vehicles older than 10 years, and 39% exceed 150,000 miles. The used car market is not a market for recent vehicles — it is dominated by aged, high-mileage inventory.	Dealerships seeking to differentiate should stock newer, lower-mileage vehicles — they represent a minority of supply but a majority of buyer intent.
4	More than 55% of sellers report their vehicle as 'excellent' condition, but price distributions for 'excellent', 'good', and 'like new' heavily overlap. Condition ratings are inflated and unreliable as a standalone signal.	Buyers should not pay a condition premium without independent verification. Condition alone should not drive a purchase decision.
5	Ford, Chevrolet, and Toyota dominate listing volume (38% combined), but volume does not equal price leadership. Ram and GMC — truck-heavy brands — show higher median prices within the top 10.	Dealerships should not conflate listing popularity with pricing power. Truck-category vehicles offer stronger margin opportunities than sedan-dominant brands at comparable ages.
6	Diesel vehicles (6.2% of listings) command measurably higher prices than equivalent gas vehicles. This premium reflects both fuel efficiency expectations and the niche supply of diesel passenger vehicles.	Buyers specifically seeking diesel should expect to pay above-market relative to comparable gas vehicles. Sellers of diesel vehicles should not underprice relative to gas comps.
7	Electric vehicles represent only 53 listings (0.19%) — statistically insufficient for any reliable EV pricing conclusion from this dataset. EV pricing cannot be generalised from this analysis.	Any stakeholder making EV inventory decisions should supplement this analysis with a dedicated EV-focused data source before acting.
8	All listings fall within months 4–5 (Spring). No seasonal price variation can be measured from this dataset. The assumption that spring listings are representative of year-round prices has not been validated.	Pricing strategies derived from this dataset should be validated against a full-year dataset before being applied to non-Spring inventory decisions.
9	Vehicles priced in the Premium category (>$15k) are disproportionately newer (under 8 years) and lower mileage. The Budget category (<$5k) is dominated by vehicles older than 15 years with over 120,000 miles.	The used car price tiers are not arbitrary — they map directly onto age and mileage bands that can be used as acquisition filters by dealerships without requiring a predictive model.
10	Automatic transmission vehicles (90.4%) dominate the market and show slightly higher median prices than manual vehicles (8.8%). However, the difference is partially confounded by vehicle type — manual vehicles skew toward smaller cars.	Transmission type alone is a weak pricing signal. Its impact should be evaluated within specific vehicle type and age segments, not across the full dataset.

12. Recommendations

Each recommendation maps directly from an insight to an action to an expected business impact.

Recommendation 1 — Sellers: Use age + mileage benchmarks before listing

Insight	Car age (-0.425) and odometer (-0.393) are the two strongest price predictors and are readily observable.
Recommendation	Before listing, sellers should compare their vehicle against others of the same age bracket (±2 years) and mileage band (±20,000 miles) using the Tableau dashboard or a similar data tool.

Expected Impact	Sellers using data-backed benchmarks are less likely to underprice by 10–15% — a $700–$1,000 improvement on a $7,000 median-priced vehicle.
Feasibility	Requires access to a pricing reference tool. The Tableau dashboard built in this project provides this at zero cost.

Recommendation 2 — Dealerships: Prioritise acquisition of vehicles aged 4–8 years with under 80,000 miles

Insight	Vehicles in the 4–8 year band represent only ~20% of listings but command mid-range to premium prices. They are underrepresented in supply but carry strong buyer demand.
Recommendation	Dealerships should set acquisition filters targeting vehicles 4–8 years old with under 80,000 miles — the segment where pricing power is strongest and depreciation curves are most predictable.
Expected Impact	Shifting even 20% of inventory acquisition toward this segment could improve average gross margin per vehicle by $800–$1,500 based on the price differential between age categories in this dataset.
Feasibility	Requires disciplined sourcing criteria. Can be implemented immediately using the data benchmarks established in this analysis.

Recommendation 3 — Buyers: Discount 'excellent' condition claims and verify against benchmarks

Insight	Over 55% of listings claim 'excellent' condition, but the price distribution overlaps heavily with 'good' and 'like new'. The condition label is unreliable without verification.
Recommendation	Buyers should treat condition as an unverified claim and base their evaluation primarily on age and odometer benchmarks. Request a third-party inspection before accepting a condition-based price premium.
Expected Impact	Avoiding a 10% overpayment on a $9,679 average vehicle saves approximately $968 per transaction. At scale, this represents significant consumer value.
Feasibility	Requires only a change in buyer evaluation process — no technology investment needed. Third-party inspections typically cost $100–$150 and are widely available.

13. Impact Estimation

Why Act Now?

The used car market has experienced above-average price volatility since 2020 due to supply chain disruptions, reduced new vehicle availability, and shifting consumer preferences toward larger vehicles. In this environment, sellers and dealerships that price accurately have a material competitive advantage. The benchmarks in this analysis are derived from actual market listings and can be applied immediately without additional investment.

Recommendation	Stakeholder	Estimated Impact	Basis
Age + mileage benchmarking before listing	Individual sellers	+$700–$1,000 per transaction (10–15% reduction in underpricing)	Based on median price of $6,750 and a conservative 10–15% underpricing assumption in benchmark-free listings
Prioritise 4–8 year acquisition for dealerships	Dealerships	+$800–$1,500 gross margin per vehicle	Based on price differential between 'Moderate' and 'Old' age categories in the cleaned dataset
Discount unverified condition claims + inspect	Individual buyers	+$900–$1,000 saved per transaction	Based on average price of $9,679 and 10% condition premium inflation estimate
Data-driven pricing tool deployment (Tableau)	Marketplace platform / dealership	3–5% reduction in days-onmarket for correctly priced listings	Based on industry benchmarks: correctly priced vehicles sell faster, reducing holding costs by $50–$150/day

14. Limitations

Data Quality Constraints

- Condition is self-reported and systematically inflated — 55% 'excellent' — reducing its reliability as a pricing predictor and potentially biasing any model that treats it as ground truth.
- Vehicle model names are highly inconsistent in the raw data (e.g. 'f-150', 'f150', 'F 150' all refer to the same model). Despite cleaning, model-level analysis may still be noisy.
- Approximately 41% of condition values required imputation using the mode — a simplification that reduces model diversity for this feature.

Scope Constraints

- The dataset covers US Craigslist listings only. Conclusions cannot be generalised to other markets, auction houses, or dealership lot pricing.
- No vehicle history data (accidents, ownership count, service records) is available. These factors are known to significantly impact resale value but cannot be captured with this dataset.
- Electric vehicle pricing cannot be analysed reliably from this dataset (only 53 EV records).

Analytical Constraints

- All listings are from months 4–5 only — seasonal trends cannot be measured or validated.

- The linear correlations reported understate the true relationship between features and price; tree-based models are expected to capture non-linear patterns that linear analysis misses.
- car_age contains one record with a value of -1 (vehicle year exceeds posting year) — a minor data quality artefact that should be filtered in the final modeling notebook.

15. Future Scope

Additional Analysis with More Data

- Incorporate a full-year dataset to enable seasonal trend analysis — currently impossible due to the Spring-only slice.
- Expand geographic analysis using zip code or county-level data to identify hyper-local price variations.
- Add vehicle history data (Carfax or AutoCheck records) to test whether accident history and ownership count are significant price predictors beyond the features already available.

New Data Sources

- Manufacturer MSRP data to compute depreciation as a percentage of original retail price — a more standardised pricing metric than raw dollar values.
- Economic indicators (regional unemployment, fuel price indices) to model how macroeconomic conditions shift used car prices over time.
- Online auction data (Manheim, ADESA) to compare dealer wholesale prices against retail listing prices in this dataset.

Predictive and Real-Time Extensions

- Deploy the Random Forest model as a lightweight API that accepts vehicle attributes and returns a predicted price range — enabling integration into marketplace listing flows.
- Build a real-time Craigslist scraper to keep the pricing benchmarks current, replacing the static snapshot dataset with a continuously updated pricing index.
- Develop an anomaly detection model that flags listings priced more than 20% above or below the predicted range — useful for buyers seeking underpriced deals and for platforms detecting fraudulent listings.

16. Conclusion

The used car pricing market suffers from information asymmetry that disadvantages all parties — buyers overpay, sellers underprice, and dealerships operate on intuition rather than data. This project addressed that gap by cleaning and analysing a dataset of 27,987 real market listings, identifying car age and odometer reading as the two dominant price drivers, and building a KPI framework and Tableau dashboard that translates those findings into actionable benchmarks for buyers, sellers, and dealerships. The core recommendation — that pricing decisions should be anchored to age-and-mileage benchmarks rather than subjective condition claims — is immediately actionable at zero additional cost, with a potential per-transaction impact of $700–$1,500. The next step is deploying a Random Forest regression model as a pricing tool and expanding the dataset to cover a full calendar year, which will enable the seasonal and regional analyses that this dataset's Spring-only scope does not permit.

17. Appendix

A. Full Data Dictionary

Column	Description	Type	Notes
id	Unique identifier for each listing	Integer	Used for duplicate removal
price	Listed vehicle price in USD	Float	Target variable; filtered $1,500–99th pct
year	Vehicle model year	Float	Filled with median; restricted to 1980+
manufacturer	Brand name	String	Unknown-filled; grouped top-10
model	Vehicle model name	String	Unknown-filled; used for smart imputation
condition	Seller-reported condition	String	Mode-filled; 55% 'excellent'
cylinders	Number of engine cylinders	Float	Extracted from text; valid: 3,4,5,6,8,10,12
fuel	Fuel type	String	Model-based imputation; 92% gas
odometer	Total miles driven	Float	Median-filled; percentilecapped
title_status	Legal title condition	String	Mode-filled
transmission	Automatic / manual / other	String	Model-based imputation; 90% automatic
drive	Drive type: fwd, rwd, 4wd	String	Model+type-based imputation
type	Body type: sedan, SUV, truck, etc.	String	Model-based imputation
paint_color	Exterior colour	String	Unknown-filled to avoid colour bias
state	US state abbreviation	String	Mode-filled
post_year	Year listing was posted (extracted from posting_date)	Float	Engineered feature
post_month	Month listing was posted	Float	Engineered; all values 4– 5 (Spring only)
car_age	post_year minus vehicle year	Float	Engineered; primary depreciation indicator
price_category	Budget / Mid-range / Premium bucket	String	Engineered for segmentation

mileage_categor y	Low / Medium / High mileage bucket	String	Engineered for segmentation
age_category	New / Moderate / Old age bucket	String	Engineered for segmentation
manufacturer_gr ouped	Top 10 manufacturers; rest = 'other'	String	Engineered to reduce chart clutter
posting_season	Spring/Summer/Fall/Winter from post_month	String	Zero variance — all Spring in this dataset

B. Cleaning Log Summary (from 02_cleaning.ipynb)

- Step 1: Loaded raw CSV — 48,471 rows × 26 columns
- Step 2: Removed duplicate rows using 'id' column
- Step 3: Converted price, year, odometer to numeric; extracted cylinders from text
- Step 4: Dropped rows with missing price (target variable)
- Step 5: Filled year and odometer with median; cylinders with mode
- Step 6: Dropped irrelevant/high-missing columns: lat, long, url, region_url, image_url, county, VIN, region, size, description
- Step 7: Applied model-based imputation for fuel, transmission, type, drive
- Step 8: Mode-filled condition, title_status, state
- Step 9: Unknown-filled manufacturer, model, paint_color
- Step 10: Filtered price outliers (<$1,500 and >99th percentile)
- Step 11: Filtered year to 1980–max; odometer to 1st–99th percentile
- Step 12: Restricted cylinders to valid values [3,4,5,6,8,10,12]
- Step 13: Converted posting_date to datetime; extracted post_year, post_month; dropped original
- Step 14: Engineered car_age, price_category, mileage_category, age_category, manufacturer_grouped, posting_season
- Final output: 27,987 rows × 23 columns — zero null values

18. Contribution Matrix

This section documents each team member's contribution across all project phases. Claims must match evidence in GitHub Insights, PR history, and committed files.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistic al Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Yaseen	-	-	✓	-	-	✓	✓
Tejaswini	✓	✓	-	-	✓	-	✓
Kashika	-	✓	-	-	-	✓	✓
Saksham	-	-	✓	✓	-	-	✓
Shubh	✓	-	-	-	✓	-	✓
Nachiket	-	✓	-	✓	-	-	✓

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artifacts.

Team Lead Signature: ______________________________________ Date: 29/04/26